Computational Methods and Complete Numerical Records

Scheduled network control — numerical methods and records

Contents

Reading and execution guide **3**

Select the appropriate execution route 3

C01. Models, metrics and distinct experiment families **4**

Deterministic network and metrics 4

Nonlinear stochastic model 4

C02. Schedules and complete deterministic comparison **5**

Table C1. Complete deterministic comparison 5

C03. Physical actuation and continuous-phase diagnostics **6**

Native-plant input and total energy 6

Continuous phase and sampled-angle unwrapping 6

C04. Contraction, graph and displacement fits **7**

Main time-history fit 7

Nine graph-rate cases 7

Displacement direction and scale fit 7

C05. Terminal-phase curve and endpoint diagnostics **8**

Original plotted curve 8

Common-deadline endpoint comparisons 8

C06. Complete primary stochastic cap study **9**

Sample design and estimands 9

Paired family and simultaneous intervals 9

Table C2. All 16 primary cap ratios 10

C07. Quadratic projection and nested refinement **10**

Fixed-design covariance-aware vertex 10

Table C3. Nested Euler–Maruyama step comparison 11

Pathwise extrapolation 11
Table C4. Pathwise Richardson diagnostic 11

C08. Numerical evaluation of analytic certificates **12**
Local initialization, metric and bound 12
Local trajectory and solver comparison 13

C09. Scalar diagnostics and precision comparisons **13**
Scalar noise sweep and modal records 13
High-precision $p = 3$ reference 13
Interpreting numerical comparisons 14

Reading and execution guide

This companion specifies the numerical models, experiments, estimators and recorded results in this snapshot. It supplies complete readable records alongside the executable code and machine-readable data. The mathematical arguments establishing control guarantees remain in the journal documents. Computational checks and numerical evaluation of certificate constants do not replace those arguments.

The package supplies scientific kernels and recorded simulation data. Tables C1–C4 give the complete comparison, cap, time-step, and extrapolation records with their units and uncertainty interpretations. Record locations maps them to the manuscript, Technical Supplement, and extended companion. Numerical equations below have local identifiers E01–E10, independent of journal numbering. C01–C09 are stable section identifiers. The data and code index maps each figure panel and table to its methods, data and code.

Read this document as PDF, as the offline HTML edition, or as Markdown. All document and data links are relative to the snapshot. No external service, authentication or network connection is needed to read the supplied PDF or HTML. The Extended Companion is provided separately in this repository; the manuscript and Technical Supplement accompany the journal submission and are not included here.

Select the appropriate execution route

Run from the repository's `simulation` directory, with the pinned numerical dependencies already installed. Each result directory must be outside the entire repository. From this directory, use `../../results`, or launch `python3 -B simulation/reproduce.py MODE --output-dir ../results` from the repository root. The output names in the table below use that external parent location. Existing outputs are not reference inputs for a different route.

Route	Command	What is newly calculated
Integrity	`python reproduce.py verify`	Checksums and exact file inventory only; standard library.
Quick	`python reproduce.py quick --output-dir ../../quick_results`	Statistics from sealed path arrays and certificate diagnostics, without new trajectories.
Targeted references	`python reproduce.py reference --output-dir ../../reference_results`	High-precision power check, physical-effort integration, deterministic protocol replay, endpoint diagnostics and covariance-aware statistics. No stochastic trajectories.
Deterministic	`python reproduce.py deterministic --output-dir ../../deterministic_results`	Figure-series/local-trajectory reconstruction, with the documented reuse of supporting records.
Same-stream replay	`python reproduce.py replay --output-dir ../../replay_results`	The first 200 primary paths and corresponding coarse paths; not independent replication.
Full	`python reproduce.py full --confirm-full --output-dir ../../full_results`	The complete distributed pipeline, including 20,000 primary and nested 10,000 stochastic paths, then the targeted stages.

Additional diagnostic and chunked-generation commands are documented in README.txt. The full route requires explicit confirmation. Partial stochastic checkpoints are not completed experiments. Resume saved numerical state only when its configuration, code, environment, and package hashes match. Start a separate output directory.

The original tested environment is CPython 3.13.5 on Linux x86_64 with NumPy 2.3.5, SciPy 1.17.0, mpmath 1.3.0, SymPy 1.14.0, Numba 0.65.1 and llvmlite 0.47.0. These are reproducibility pins, not claims of latest versions. Dependency installation is external to execution and can require network access. The commands do not install dependencies. The PDF/HTML are already rendered and numerical execution needs no typesetting software. Cross-platform bitwise equality is not assumed.

C01. Models, metrics and distinct experiment families

Deterministic network and metrics

There are $N = 6$ planar nodes. Let $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and

$$f(x) = \alpha x + \omega J x - b_{\mathrm{SL}} \|x\|_2^2 x, \qquad \alpha = b_{\mathrm{SL}} = 1, \quad \omega = 2.$$

Let $X = \mathrm{col}(x_1, \dots, x_N)$, $F(X) = \mathrm{col}(f(x_1), \dots, f(x_N))$, and let L be the unit-weight undirected graph Laplacian. Define the arithmetic mean $y = N^{-1} \sum_i x_i$ and the stacked disagreement $\eta = X - \mathbf{1} \otimes y$. The network horizon is $T = 2$. Except where a graph-specific gain is specified in C04, the coupling multiplier is $k_0 = 1.2$ and the graph is the six-node ring. The reduced reference $\bar{y}$ solves $\dot{\bar{y}} = f(\bar{y})$ from the arithmetic initial mean. The snapshot is $t_\star = T - 10^{-5} = 1.99999$.

The fiber-only deterministic model is

$$\dot{X} = F(X) - k_0 \kappa(t)(L \otimes I_2)X. \tag{E01}$$

Reported disagreement is $\|\eta\|_2$, not normalized by N. Projection-orbit error is $e_y(t) = \|y(t) - \bar{y}(t)\|_2$. Phase error is the continuously tracked angular difference of the mean relative to the reduced reference, in radians. Coupling energy is $J_u = \int_0^{t_\star} \| - k_0 \kappa (L \otimes I_2) X\|_2^2 dt$. The full-clock diagnostic requires additional physical input, accounted for in C03. Time and state scales follow the numerical benchmark; no physical-unit conversion is specified in the source records.

The finite-amplitude Cartesian array is defined, in node order, by

$$x_i^0 = r_i(\cos\theta_i, \sin\theta_i)^\top,$$
$$r = (0.75, 1.25, 0.90, 1.15, 0.80, 1.20), \qquad \theta = (-0.75, -0.42, -0.15, 0.18, 0.48, 0.82).$$

The main time histories assign this state at $t = 0$. The graph-rate tests assign the same array directly at $t_0 = 1.6$. They do not use the state obtained by advancing the main time history to 1.6. The displacement state in C04, the unit-radius stochastic state below, and the local-certified state in C08 are separate initializations.

Nonlinear stochastic model

Let $W = (W_1, \dots, W_N)^\top$ consist of independent standard node Brownian motions and let $e_\theta(t) = (-\sin\omega t, \cos\omega t)^\top$. The implemented equation is

$$\begin{aligned} dX = {} & [F(X) - k_0 \kappa_K(t)(L \otimes I_2)X]dt \\ & - k_0 \sigma \kappa_K(t)(L \otimes e_\theta(t))dW_t, \qquad \kappa_K(t) = \min\{(T-t)^{-1}, K\}. \end{aligned} \tag{E02}$$

The Laplacian acts on node measurement channels, not independent edge channels. The reference tangent is common and deterministic, not a tangent computed from each node or the current mean. Its time-dependent, state-independent diffusion has zero Itô–Stratonovich correction. The numerical rule is left-point Euler–Maruyama. Although $\mathbf{1}^\top L = 0$, nonlinear drift can transmit disagreement effects to the mean.

The nonlinear initial state has unit radii and centered phases

$$x_i(0) = (\cos\vartheta_i, \sin\vartheta_i)^\top, \qquad \vartheta_i = 2(a_i - \bar{a}),$$
$$a = (0.08, -0.05, 0.12, -0.10, 0.03, -0.08)^\top, \qquad \bar{a} = N^{-1}\sum_i a_i.$$

For a unit eigenvector v_j of L, the scalar comparison starts from $\xi_{j0} = v_j^\top(\vartheta - \bar{\vartheta}\mathbf{1})$. These are phase offsets (arc lengths on the specified unit cycle), not Cartesian sine projections. In co-rotating radial/tangent coordinates, the node variational rates are -2α and zero. Nonzero graph modes are tangent to the node cycle and normal to synchronization. Their scalar comparison is

$$d\xi_j = -\rho_j\kappa_K(t)\xi_j dt - \rho_j\sigma\kappa_K(t)d\widetilde{W}_j(t), \qquad \rho_j = k_0\lambda_j(L).$$

Here $\widetilde{W}_j$ is the orthogonally transformed Brownian coordinate. This is a variational model, not an exact nonlinear stochastic reduction. The nonlinear terminal cost is squared Euclidean disagreement, $\sum_i \|x_i(T) - y(T)\|_2^2$, rather than phase variance. The scalar/modal comparator is evaluated with its declared phase-offset initialization.

Source records. Model, experiment configuration, network implementation, primary stochastic kernel.

C02. Schedules and complete deterministic comparison

The four fiber-only profiles are $\kappa_0 = 1/T$, $\kappa_1 = (T-t)^{-1}$, $\kappa_2 = T/(T-t)^2$ and the capped inverse-time profile. All use initial coupling coefficient k_0/T. The cap is fixed at the stochastic comparator's $K_{\text{opt}} = 2.53979193583334$. The full-clock row instead uses $\dot{X} = \kappa_1[F(X) - k_0(L \otimes I_2)X]$ and accelerates the node drift as well as coupling.

For interpreting the functional columns, define $E_\kappa(t) = \exp[-\rho\int_0^t \kappa(u)du]$, $I_\kappa^{(n)} = \int_0^T E_\kappa(t)^n dt$, $I_\kappa = I_\kappa^{(1)}$ and $\rho = 1.2$. These columns record the integrals used in the comparison, not an empirical fit. The second-order column corresponds to the node-average cancellation in the Stuart–Landau model. Its mathematical justification remains in the journal treatment.

The label "Xu–Liu ($\ell = 2$)" denotes the restricted single-weight, symmetric regulatory coupling. Setting $C(t) = (T-t)^2$ and $\bar{k} = k_0 T$ gives coefficient $\bar{k}/C(t) = k_0\kappa_2(t)$. This row does not implement the complete directed, multiweighted controller associated with that name.

Table C1. Complete deterministic comparison

All rows use the finite-amplitude state and common snapshot $t_\star$. The phase column is in radians. The inverse-square disagreement is reported at the 10^{-14} numerical-resolution threshold. J_u counts coupling energy only, including in the full-clock row. The cap is displayed at its original six-decimal rounding.

Profile	I_κ/T	$I_\kappa^{(2)}/T$	$\|\eta(t_\star)\|$	e_y	Phase	J_u
Constant	0.582338	0.378868	3.4102×10^{-1}	2.1966×10^{-2}	-3.64×10^{-3}	0.865842
Inverse time	0.454545	0.294118	4.9802×10^{-7}	1.0305×10^{-2}	-3.35×10^{-3}	1.235583
Xu–Liu ($\ell=2$)	0.368879	0.247595	$<10^{-14}$	7.1092×10^{-3}	-3.16×10^{-3}	1.466261
Cap $K = 2.539792$	0.458124	0.294455	4.9009×10^{-2}	1.0409×10^{-2}	-3.35×10^{-3}	1.203441
Full clock	–	–	5.0472×10^{-7}	1.4064	2.04121×10^{1}	1.259392

The four fiber-only sampled reduced-flow errors have the same ordering as $I_\kappa^{(2)}/T$. This is consistent with the displacement bound and does not establish finite-amplitude proportionality. Inverse square

has the smallest listed functionals and disagreement, but the largest coupling energy. Schedule ordering does not order actuation effort.

Record links. Comparison metrics, complete method metrics, functional integrals. The deterministic route regenerates the displayed-series view with its documented supporting-record reuse; the targeted route separately checks selected protocols. Full regeneration is a distinct, explicitly confirmed route.

C03. Physical actuation and continuous-phase diagnostics

Native-plant input and total energy

For the native plant $\dot{X} = F(X) + u$, the full-clock realization requires

$$u_c = -k_0\kappa_1(L \otimes I_2)X, \qquad u_w = (\kappa_1 - 1)F(X), \qquad u = u_c + u_w.$$

With $J_w = \int_0^{t_\star} \|u_w\|_2^2 dt$, the physical energy is

$$J_{\text{total}} = J_u + J_w + J_{\text{cross}}, \qquad J_{\text{cross}} = 2\int_0^{t_\star} u_c^\top u_w dt. \tag{E03}$$

It is accumulated both by components and directly from $\|u_c + u_w\|_2^2$. Omitting the cross term is not the total-input calculation. The normalization is exactly $\kappa_1 = (T-t)^{-1}$. In the regular variable $\tau = \log[T/(T-t)]$, the physical-time Jacobian is $dt/d\tau = Te^{-\tau}$.

The finite-amplitude DOP853 integration uses relative/absolute tolerance pairs $(10^{-10}, 10^{-12})$, $(10^{-12}, 10^{-13})$ and $(2.3 \times 10^{-14}, 10^{-15})$. The finest stored result is

Quantity	Stored finest result
J_u	1.2593922836984972
J_w	2399448.0144150066
J_{cross}	−0.10973109273849226
J_{total}	2399449.164076198
Continuous phase excess	20.41208384685857 rad

The binary64 cutoff is the same $t_\star = 1.99999$ used by the original source, for which $T - t_\star = 1.0000000000065512 \times 10^{-5}$. Direct and component accumulation agree within the stated numerical diagnostic tolerance. These are integration-consistency results, not rigorous integration-error bounds or an equal-total-actuation comparison.

For the synchronized nonstationary cycle, $\|F(X)\|_2^2 = N\omega^2 a_\star^2 = 24$ and the diagnostic extra-energy expression is

$$24[1/\delta - 1/T - 2\log(T/\delta) + (T-\delta)], \qquad \delta = T - t_\star.$$

Its leading term is $24/\delta$, without an extra T^2 under this normalization. The finite-amplitude numbers above are directly integrated, not replaced by that cycle expression. The separate cutoff sequence uses nominal $\delta = 10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}$.

Continuous phase and sampled-angle unwrapping

At the common snapshot, $\tau_\star \simeq 12.2060726455$ and continuous accumulation gives phase excess 20.4120838469 rad. The reduced-cycle expression $\omega\{\log[T/(T-t_\star)] - t_\star\} = 20.4121652910$ rad differs by the finite-disagreement transient 8.14×10^{-5} rad.

On 4001 uniform physical times in $[0, t_\star]$, `numpy.unwrap` applied only to sampled angles gives 14.1288985397 rad. The final continuous increment is 7.86364 rad, and this grid misses one revolution, 6.28318530718 rad. This observation is specific to the given output grid, not a universal sample-count threshold. The snapshot is not a finite terminal phase of the warped dynamics.

Record links. Physical-energy snapshot, cutoff series, phase-grid check, targeted effort implementation.

C04. Contraction, graph and displacement fits

All quoted slopes are finite-window least-squares fits of log error against log scale. They do not certify an asymptotic exponent or integration error. The main time-history, graph and displacement simulations use $T = 2$, $\kappa_1(t) = (T - t)^{-1}$ and physical-time Radau with relative/absolute tolerances $10^{-10}/10^{-12}$.

Main time-history fit

The finite-amplitude ring state is assigned at $t = 0$, with $k_0 = 1.2$. There are 4001 uniformly spaced output times through $t_\star$. For $s = (T - t)/T$, the mask $s < 0.03$ selects 120 samples, giving the original stored exponent 1.1996984425895174, reported as **1.1997** against $\rho = 1.2$. The displayed time-series file retains every fourth sample and every sample among the final 81, giving 1061 unique displayed rows. Display decimation does not change the fit population.

Nine graph-rate cases

At $t_0 = 1.6$, assign the finite-amplitude Cartesian array in C01 directly, in the same node order, to each six-node unit-weight path, ring or complete graph. Their spectral gaps are $2 - \sqrt{3}$, 1 and 6. For each target $\rho \in \{0.6, 1, 1.4\}$, use $k_0 = \rho/\lambda_2(L)$.

The full parent grid is

$$s_j = 0.2(10^{-4})^{j/499}, \qquad t_j = 2(1 - s_j), \qquad j = 0, \dots, 499. \tag{E04}$$

The implementation is `np.geomspace(2e-5, 0.2, 500)[::-1]`. The mask $2 \times 10^{-4} < s_j < 0.03$ with $\|\eta(t_j)\|_2 > 10^{-12}$ selects 272 points in each case. These are zero-based indices 103–374, with extreme s values approximately 0.0002009250 and 0.0298798693.

Graph	Target 0.6	Target 1.0	Target 1.4
Path	0.5985491	0.9985998	1.3986662
Ring	0.5992636	0.9991330	1.3991638
Complete	0.5991917	0.9993635	1.3994543

The maximum recorded absolute target difference is 0.001450911842219904, displayed as 1.451×10^{-3}. Digits identify stored fits. This is a directly initialized near-deadline rate test, not a full-history comparison from $t = 0$.

Displacement direction and scale fit

The ring uses $k_0 = 1.2$ and an initialization separate from the local-certified run,

$$y_b = 0.8(\cos 0.2, \sin 0.2)^\top, \qquad x_i(0) = y_b + \epsilon D_{i,:}^\top,$$

$$D = \frac{1}{24}\begin{pmatrix} 10 & 2 \\ -7 & 5 \\ 3 & -11 \\ -5 & -4 \\ 8 & 9 \\ -9 & -1 \end{pmatrix}, \qquad \mathbf{1}^\top D = 0, \qquad \|D\|_F = 1. \tag{E05}$$

Thus $\epsilon = \|\eta(0)\|_2$. The exact matrix specifies the existing binary64 direction computed by column-centering and Frobenius normalization; no saved initialization is replaced. Use $\epsilon_j = 0.015\,40^{j/11}$ for $j = 0, \dots, 11$, 1601 uniform times through $t_\star$, and the first eight scales in the fit. The eighth scale is approximately 0.1568859243. The reduced reference uses DOP853 at tolerances $10^{-12}/10^{-13}$. The original terminal projection-orbit slope is 1.9761166019943055, displayed as **1.9761**. This finite-window value depends on the specified direction and is consistent with quadratic order.

Record links. Original fitted graph rates, displacement curve, original summary, targeted graph replay, targeted displacement replay, protocol implementation. Original fit values and later recalculated values retain separate fields and filenames.

C05. Terminal-phase curve and endpoint diagnostics

Original plotted curve

The phase experiment uses the finite-amplitude ring state at $t = 0$, $T = 2$, $k_0 = 1.2$ and κ_1. Full-state DOP853 integrates in $\tau = -\log s$ with relative/absolute tolerances $2 \times 10^{-12}/2 \times 10^{-13}$. The raw mean y_{30} at $\tau = 30$ defines the plotted reduced orbit $\phi_{t-T}(y_{30})$.

The parent grid is

$$s_j = 0.2(1/400)^{j/239}, \quad j = 0, \dots, 239, \qquad \texttt{np.geomspace(0.2,5e-4,240).} \tag{E06}$$

The mask $10^{-3} \le s_j \le 0.05$, with phase-tail error above 5×10^{-13}, selects 156 points, indices 56–211. Their extreme values are approximately 0.0010088204 and 0.0491294949. They are a subset of the parent grid, not a newly generated 156-point grid. The original network slope is 3.4072924626261014, displayed as **3.4073** against $2\rho + 1 = 3.4$. The sharp scalar reference $Ts^{\rho+1}/(\rho+1)$ has analytical exponent 2.2; it is evaluated directly rather than separately integrated.

Common-deadline endpoint comparisons

Transport finite-time numerical means to the same physical deadline by

$$\widehat{y}_T^{(\tau)} = \phi_{Te^{-\tau}}\big(y_{\mathrm{num}}(T(1 - e^{-\tau}))\big)\,.$$

The raw $\tau = 25, 30$ means differ by approximately 5.45×10^{-11}, which includes remaining reduced motion. The smallest fitted network tail is approximately 8.3×10^{-12}. The reduced-flow round trip at one estimated point is 3.14×10^{-16}; that composition error is not the true error of the terminal estimate.

Five full-state/scaled-normal refinements compare endpoints at $\tau = 25, 30, 35$. In the stored runs, transported endpoints agree within 10^{-15} within each run, and the largest between-run separation is $1.1969568733894625 \times 10^{-13}$. Transported-tail slopes range from 3.407363944702072 to 3.407767611106687. The full-state refinements use tolerances $10^{-13}/10^{-14}$ with maximum step 0.1, and $2.3 \times 10^{-14}/10^{-15}$ with maximum step 0.05, together with corresponding mean/scaled-normal formulations. The original setting is retained separately.

Thirty deterministic endpoint perturbations, using plus/minus 10^{-12} along the coordinate axes and diagonal in each run, record sensitivity. These are consistency diagnostics, not random replications, validated endpoint-error bounds or extra certified slope digits. The original plotted curve is not replaced by a transported-endpoint curve.

Record links. Original curve, identical plotted-series record, targeted endpoint results, sensitivity probes, endpoint implementation.

C06. Complete primary stochastic cap study

Sample design and estimands

For $\sigma = 0.002$, the scalar-mode calculation gives the fixed reference cap $K_{\mathrm{opt}} = 2.53979193583334$. All 16 cap ratios listed in Table C2 use the same Brownian increments on each path. There are 20,000 independent primary paths at fine step $h_{\mathrm{f}} = 5 \times 10^{-4}$. The first 10,000 also carry coarse step $h_{\mathrm{c}} = 10^{-3}$ at ratios $0.95, 1, 1.05$, summing adjacent fine increments. Their fine terminal values are exactly the corresponding primary subset. This is 20,000, not 30,000, independent paths. NumPy PCG64 uses seed 20260829, batch size 200 and the preserved ordering.

Here h is the Euler–Maruyama time step (the journal's Δt). For path i, define $Z_{i,h}(K) = \sum_j \|x_j(T) - y(T)\|_2^2$ and $M_h(K) = \mathbb{E} Z_{i,h}(K)$. For n independent paths,

$$\widehat{M}_h(K) = \frac{1}{n} \sum_i Z_{i,h}(K), \qquad \widehat{\mathrm{se}}_h(K) = \left\{ \frac{\sum_i [Z_{i,h}(K) - \widehat{M}_h(K)]^2}{n(n-1)} \right\}^{1/2}. \tag{E07}$$

The marginal 95% interval is $\widehat{M}_h \pm t_{n-1,0.975} \widehat{\mathrm{se}}_h$, where $t_{\nu,p}$ is a Student-t quantile. This is large-sample calibration for the fixed-step mean, without assuming Gaussian squared disagreements. It does not quantify continuous-time accuracy.

For the exact scalar variational prediction $M_{\mathrm{mod}}(K)$ with the phase-offset initialization, the point discrepancy is $\delta = 100[\widehat{M}_{h_{\mathrm{f}}}(K)/M_{\mathrm{mod}}(K) - 1]$. Its marginal interval is the same affine transformation of the interval for $M_{h_{\mathrm{f}}}$.

Paired family and simultaneous intervals

Define $D_i(K) = Z_{i,h_{\mathrm{f}}}(K) - Z_{i,h_{\mathrm{f}}}(K_{\mathrm{opt}})$. The family has 15 nonzero contrasts against the fixed analytical reference, not all pairwise cap comparisons. With $\alpha_{\mathrm{sig}} = 0.05$,

$$\bar{D}(K) \;\pm\; t_{n-1,1-\alpha_{\mathrm{sig}}/(2\cdot 15)} \left\{ \frac{\sum_i [D_i(K) - \bar{D}(K)]^2}{n(n-1)} \right\}^{1/2}, \qquad n = 20000. \tag{E08}$$

This retains the common-random-number covariance. The critical value is 2.9355523, with the same large-sample qualification. Over $[0.95, 1.05]K_{\mathrm{opt}}$, the minimum reference-output correlation is 0.998707 and the minimum variance-reduction factor for a paired difference versus independent sampling is 557.2.

Table C2. All 16 primary cap ratios

Modal and nonlinear means are multiplied by 10^4. The ± values are **marginal 95% half-widths**, not standard errors. δ is percent of the modal mean-square prediction. $\bar{D}$ and its **Bonferroni-adjusted simultaneous 95% interval** are multiplied by 10^8. The reference contrast is exactly zero.

K/K_{opt}	Modal	Nonlinear ± marginal half-width	δ	$\bar{D}$	Simultaneous interval
0.600	1.470770	1.477649 ± 0.006838	+0.468	+4415.221	$[+4354.876, +4475.566]$
0.750	1.151298	1.155969 ± 0.007036	+0.406	+1198.413	$[+1161.834, +1234.991]$
0.850	1.067090	1.071383 ± 0.007374	+0.402	+352.557	$[+330.920, +374.194]$
0.900	1.046052	1.050342 ± 0.007585	+0.410	+142.150	$[+127.815, +156.485]$
0.950	1.034915	1.039296 ± 0.007818	+0.423	+31.690	$[+24.561, +38.819]$
0.975	1.032380	1.036836 ± 0.007942	+0.432	+7.091	$[+3.536, +10.646]$
0.990	1.031703	1.036214 ± 0.008018	+0.437	+0.867	$[-0.553, +2.287]$
0.995	1.031608	1.036139 ± 0.008044	+0.439	+0.113	$[-0.596, +0.823]$
1.000	1.031577	1.036127 ± 0.008070	+0.441	+0.000	$[+0.000, +0.000]$
1.005	1.031608	1.036179 ± 0.008096	+0.443	+0.516	$[-0.193, +1.225]$
1.010	1.031699	1.036292 ± 0.008122	+0.445	+1.642	$[+0.225, +3.060]$
1.025	1.032323	1.036982 ± 0.008202	+0.451	+8.545	$[+5.007, +12.083]$
1.050	1.034459	1.039238 ± 0.008337	+0.462	+31.104	$[+24.046, +38.163]$
1.100	1.042358	1.047411 ± 0.008618	+0.485	+112.835	$[+98.784, +126.886]$
1.250	1.087813	1.093838 ± 0.009517	+0.554	+577.103	$[+542.390, +611.815]$
1.500	1.206916	1.214953 ± 0.011142	+0.666	+1788.257	$[+1719.907, +1856.608]$

Three nonzero contrasts, at ratios $0.990, 0.995, 1.005$, remain unresolved relative to the reference; including the exact reference gives four displayed caps. Every other contrast has a positive lower endpoint. No tested cap is significantly better. This fixed-grid comparison neither supplies a continuous-domain optimizer interval nor establishes equal expected mean-square errors.

At the reference, the relative standard error is 0.3991% of modal mean-square error and 0.3974% of the sample mean. The marginal half/full widths are 0.7823%/1.5646% of the modal value. The point discrepancy 0.4411% has interval $[-0.3412\%, 1.2234\%]$. The largest absolute point discrepancy is 0.6659% at ratio 1.5, with interval $[-0.2573\%, 1.5891\%]$. Sample maxima are not uniform nonlinear–modal error bounds.

Record links. Sealed pathwise arrays, full-precision cap statistics, primary summary, table formatter. The exploratory `network_noise_validation.csv` and its maximum discrepancy in the older `numerical_summary.json` are separate records, not the 20,000-path primary cap result. Table C2 is reconstructed from the sealed primary NPZ and its reporting files.

C07. Quadratic projection and nested refinement

Fixed-design covariance-aware vertex

Use $x = K/K_{\mathrm{opt}} - 1$ on the seven ratios $\{0.975, 0.990, 0.995, 1, 1.005, 1.010, 1.025\}$. Let the design matrix X have rows $(1, x, x^2)$ and let Z_i collect each path's seven outcomes. The pathwise coefficients are $b_i = (\mathsf{X}^\top \mathsf{X})^{-1} \mathsf{X}^\top Z_i$; the independent reference implementation evaluates the equivalent least-squares problem by QR. Let $\bar{b} = n^{-1} \sum_i b_i$.

The vertex is $\widehat{K}_{\mathrm{quad}} = K_{\mathrm{opt}}[1 - \bar{b}_1/(2\bar{b}_2)]$. Its delta-method standard error uses the sample covariance S_b of the pathwise coefficient vectors, divided by n, and gradient

$$g_b = \left(0, -\frac{K_{\mathrm{opt}}}{2\bar{b}_2}, \frac{K_{\mathrm{opt}}\bar{b}_1}{2\bar{b}_2^2}\right)^\top, \qquad \widehat{\mathrm{se}}(\widehat{K}_{\mathrm{quad}}) = \sqrt{g_b^\top (S_b/n) g_b}.$$

Thus cross-cap covariance is retained. The reported values are

$$\widehat{K}_{\mathrm{quad}} = 2.5366693, \qquad \widehat{\mathrm{se}} = 0.00490771, \qquad [2.5270498, 2.5462888] \quad (95\%).$$

The interval targets the seven-point quadratic projection of the fixed-step mean curve. Its inclusion of K_{opt} and $R^2 = 0.999728$ remove neither quadratic-model error, discretization bias nor off-grid uncertainty. The archived reporting and independently reconstructed last digits remain distinct numerical records.

Table C3. Nested Euler–Maruyama step comparison

This table uses the first 10,000 primary paths. Means are multiplied by 10^4. Relative coarse-minus-fine shift uses the **fine subset mean** as denominator. The final column is the paired **marginal 95% interval** for $M_{h_{\mathrm{c}}} - M_{h_{\mathrm{f}}}$, multiplied by 10^8.

K/K_{opt}	$h_{\mathrm{c}} = 10^{-3}$	$h_{\mathrm{f}} = 5 \times 10^{-4}$	Relative shift	Paired interval
0.95	1.037529	1.036400	0.1089%	$[10.8424, 11.7384]$
1.00	1.034344	1.033039	0.1262%	$[12.5438, 13.5370]$
1.05	1.037463	1.035975	0.1436%	$[14.3272, 15.4229]$

At the reference, the fine subset mean is 1.033039×10^{-4}, whereas the full 20,000-path fine mean is 1.036127×10^{-4}. Their difference is not a time-step effect. Coarse-minus-fine shifts are 0.1089–0.1436% of fine subset means. Paired intervals exclude zero and both steps preserve the three-cap ordering, without bounding remaining discretization bias.

Pathwise extrapolation

Form the covariance-preserving diagnostic

$$Z_i^{\mathrm{R}}(K) = 2Z_{i,h_{\mathrm{f}}}(K) - Z_{i,h_{\mathrm{c}}}(K), \qquad \widehat{M}_{\mathrm{R}} = \frac{1}{n_{\mathrm{r}}} \sum_i Z_i^{\mathrm{R}}, \quad n_{\mathrm{r}} = 10000. \tag{E09}$$

If $s_{\mathrm{f}}^2, s_{\mathrm{c}}^2$ and $s_{\mathrm{f,c}}$ denote the sample variances and covariance on those same paths, centering Z_i^{R} gives $2(Z_{i,h_{\mathrm{f}}} - \widehat{M}_{h_{\mathrm{f}}}) - (Z_{i,h_{\mathrm{c}}} - \widehat{M}_{h_{\mathrm{c}}})$. Squaring and summing yields

$$n_{\mathrm{r}} \widehat{\mathrm{se}}_{\mathrm{R}}^2 = 4s_{\mathrm{f}}^2 + s_{\mathrm{c}}^2 - 4s_{\mathrm{f,c}}. \tag{E10}$$

The interval targets $2M_{h_{\mathrm{f}}} - M_{h_{\mathrm{c}}}$. Removal of first-order weak bias would additionally require an expansion $M_h = M_0 + c_1 h + O(h^2)$, which is not established by two step sizes.

Table C4. Pathwise Richardson diagnostic

The extrapolated mean and standard error are multiplied by 10^4. The discrepancy δ_{R} and its marginal 95% interval are in percent of the modal prediction. Intervals quantify sampling uncertainty of the extrapolated expectation, not residual discretization bias.

K/K_{opt}	$\widehat{M}_{\mathrm{R}}$	Standard error	δ_{R}	Marginal interval
0.95	1.035271	0.005649	+0.0344	$[-1.0355, +1.1043]$
1.00	1.031735	0.005826	+0.0154	$[-1.0917, +1.1224]$
1.05	1.034488	0.006015	+0.0028	$[-1.1370, +1.1425]$

The discrepancies are $0.0344\%, 0.0154\%, 0.0028\%$, with marginal half-widths $1.0699\%, 1.1071\%, 1.1398\%$. Sampling-precision consistency with the modal prediction does not establish 0.04% agreement or isolate a bias-limited residual. Sampling error, discretization and nonlinear–modal mismatch all contribute. The paired comparisons, quadratic projection and extrapolation reuse paths and are not independent confirmations.

Record links. Quadratic-fit record, step comparison, sample split, Richardson statistics, independent statistical reconstruction.

C08. Numerical evaluation of analytic certificates

This section records numerical parameters, bound evaluations and trajectory diagnostics. The analytic construction and its hypotheses remain in the mathematical supplement. The finite-amplitude comparison has initial disagreement approximately 1.3217504433 and is not relabeled as satisfying the local gate.

Local initialization, metric and bound

Let $y_0 = N^{-1} \sum_i x_i^0$ for the finite-amplitude array in C01. Its radius is 0.8746881126. Preserve that mean and set

$$x_i^{\mathrm{loc}}(0) = y_0 + 0.01 \frac{x_i^0 - y_0}{\left(\sum_j \|x_j^0 - y_0\|_2^2 \right)^{1/2}}.$$

This is the mean-preserving local state, not the independent direction in C04. Its disagreement is $q_0 = 0.01$. In the certificate's product metric, $\|dX\|_*^2 = \|dy\|_2^2 + \|d\eta\|_2^2$, so the normal norm equals Euclidean disagreement and the base norm equals Euclidean mean norm.

The annular base sets use radii $[0.87, 0.90]$ and $[0.85, 1.002]$, with extended disk radius 1.15, normal tube radius $q_* = 0.07$, $T = 2$ and $\rho = 1.2$. Numerical evaluation of the accepted certificate yields the following selected values. Full 70-digit values are stored from an 80-digit calculation.

Evaluated quantity	Numerical value
ρ	1.2
$\beta = \mu_g$	0.2775
$c_r = \gamma$	3.076
L_2	0.512666666667
L_h	0.0358866666667
b_*	0.49282
d_*	0.00414123347845
Normal gate	0.0261240211607
Base gate	0.0271955489791
Maximum backward radius	1.13063963675
Backward collar margin	0.0193603632459
$\mu_{g,-}$	2.9675
Order-three gap with $m = 4$	0.45

These constants are an evaluated certificate, not a fitted model. The resulting comparison curve is

$$q_{\mathrm{bound}}(t) = 0.01 e^{0.49282 t} \left(\frac{2 - t}{2} \right)^{1.2}.$$

The two evaluated no-exit bounds are $0.026795262325 < 0.07$ and $0.001522761493 < 0.004141233478$. Their analytic derivation, the backward collar and the relation to the main first-exit result are retained

in the journal documents, not supplied exclusively by code.

The ball-based domain is distinct, with normal radius 2 and much smaller admitted initialization. Its displayed gate branches are approximately 5.45×10^{-10} and 1.01×10^{-13}. Its polynomial/radial calculations remain journal mathematics. The certificate diagnostic modules retain the numerical evaluations of both domain choices; the larger annular gate is not retroactively applied to the finite-amplitude initial state.

Local trajectory and solver comparison

The selected run evolves the mean and cancellation-free scaled normal coordinates $e^{\rho\tau}(V_\perp^\top \otimes I_2)\eta$ in $\tau = \log[T/(T-t)]$. It uses DOP853 with tolerances $10^{-12}/10^{-14}$, maximum τ step 0.1, and 1000 output times through $t_\star$. Refinement to $10^{-13}/10^{-15}$ and maximum step 0.05 changes sampled disagreement by at most 1.61×10^{-12} relatively. Independent full-state integrations in regular and physical time differ from the refined disagreement by at most 4.38×10^{-8} and approximately 9.29×10^{-6} relatively. The full-precision stored physical-time diagnostic is $9.2824490939 \times 10^{-6}$; the latter displayed value is the retained journal approximation.

At the snapshot, $q(t_\star) \simeq 3.845727779 \times 10^{-9}$ and $q_{\mathrm{bound}}(t_\star) \simeq 1.166325788 \times 10^{-8}$. The sampled mean radius spans $[0.8746881126, 0.9971992474]$. The maximum sampled ratio q/q_{bound} is 1, attained initially. These are solver-consistency and sampling records. The plots do not prove continuous-time invariance or exact contact at the unintegrated deadline.

Record links. Certificate constants, local states, solver comparisons, trajectory, local summary, certificate diagnostic, high-precision reconstruction.

C09. Scalar diagnostics and precision comparisons

Scalar noise sweep and modal records

The scalar comparison uses $(\eta_0, \rho, T) = (1, 1.2, 1)$ and the 31-point grid $\sigma_j = 10^{-9+j/5}$, $j = 0, \ldots, 30$, distinct from the network horizon $T = 2$. The least-squares log root-mean-square-error/log noise slopes are 0.934135, 0.934224 and 0.931788 for the exact, refined and leading inverse-square curves. The exact bounded-profile benchmark gives 0.999999990; the inverse-time fit is approximately 0.705882. All optimized named-family points on the specified sweep lie above the benchmark.

These are finite-grid fits, not new asymptotic exponents. The code evaluates accepted scalar moments and refinements; their main moment and cap results are in the manuscript and Technical Supplement; local derivative and power-family refinements are in the Extended Companion. Scalar deterministic expansions are distinct from the network sampling intervals. Modal leading-rate diagnostics, degeneracies and bounded deterministic-measurement limits have separate numerical records and are not substituted for nonlinear stochastic evidence.

Records. Scalar slope diagnostics, noise optima, exact benchmark curve, modal diagnostics, bounded-noise diagnostics.

High-precision $p = 3$ reference

The normalized profile is $\kappa_p(t) = T^{p-1}/(T-t)^p$, capped at K. For the numerical precision check, $(p, \eta_0, \rho, T) = (3, 1, 1.2, 1)$ and $\sigma \in \{10^{-5}, 10^{-8}, 10^{-12}\}$. The exact objective is evaluated by split integration of its scalar transition kernel. Its derivative root is calculated at both 60 and 80 decimal digits, with quadrature versus an independent antiderivative, independent differentiation, neighboring derivative-sign checks and finite objective probes.

σ	$\lvert K_{\mathrm{L}}/K_{\mathrm{ref}} - 1\rvert$
10^{-5}	$7.8463208697 \times 10^{-9}$
10^{-8}	$6.1369645291 \times 10^{-13}$
10^{-12}	$2.2541369701 \times 10^{-18}$

The archived `asymptotic_power_p3_accuracy.csv` and its identical reference copy were produced by double-precision bounded minimization. The high-precision root calculation supersedes only the precision interpretation of that optimizer diagnostic. It does not replace a theorem, stochastic outcome, plotted phase curve or original cap study. The reference calculation is not a new proof of global optimality or a sharper asymptotic remainder.

Records. Original optimizer record, identical archived copy, high-precision successor, full precision checks, precision implementation, supersession map.

Interpreting numerical comparisons

Table reconstruction, same-stream regeneration and independent replication are different claims. Existing numerical comparisons report their tolerances and exact equality separately; near-zero absolute agreement can coexist with large relative discrepancies.

The landing page, data and code index, record-location map and package description provide local navigation. The manuscript and Technical Supplement are submitted separately and are not distributed here. Their result and equation numbers are used as printed scholarly references. This package grants no additional redistribution license.